EXP 20: Implement Future Sales Prediction using a suitable machine learning algorithm

CODING:

```
# Future Sales Prediction using Linear Regression

# Historical sales data
months = [1, 2, 3, 4, 5, 6]
sales = [10000, 12000, 14000, 16000, 18000, 20000]

n = len(months)

# Calculate mean
mean_x = sum(months) / n
mean_y = sum(sales) / n

# Calculate slope
numerator = 0
denominator = 0

for i in range(n):
    numerator += (months[i] - mean_x) * (sales[i] - mean_y)
    denominator += (months[i] - mean_x) ** 2

slope = numerator / denominator

# Calculate intercept
intercept = mean_y - slope * mean_x
```

```
# Predict future sales

future_month = 7

predicted_sales = slope * future_month + intercept


print("FUTURE SALES PREDICTION")

print("-----------------------")

print("Month:", future_month)

print("Predicted Sales: Rs.", round(predicted_sales))
```

OUTPUT:

